

Conditional diffusion models as generative surrogates for reservoir computing substrates

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Physical reservoir computing (RC) [1–5] exploits the transient dynamics of a nonlinear system as a computational substrate, training only a linear readout [6]. Spintronic systems—such as single magnetic domain walls (DWs) [7] and interconnected nanomagnetic ring arrays [8, 9] of interest in our group—are promising candidates [4]. Characterising and optimising such a substrate requires sampling its trajectory distribution across many device configurations; yet a governing equation is rarely available for fabricated devices, and direct measurement is costly. We learn a ‘digital twin’ or surrogate for this distribution from observed trajectories alone using a conditional denoising diffusion probabilistic model [10].

We seek a generative sampler $g_\phi : (\mathbf{c}, \mathbf{z}) \mapsto \mathbf{q}, \mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, drawing trajectories $\mathbf{q} \in \mathbb{R}^{N_T}$ from the underlying data distribution $p(\mathbf{q} | \mathbf{c})$; the conditioning $\mathbf{c}$ may be a scalar, a parameter vector, or a time-dependent driving waveform. Clean trajectories $\mathbf{q}_0$ are corrupted toward Gaussian noise over N_K steps; a 1D conditional U-Net [11] $\eta_\phi(\mathbf{q}_k, k, \mathbf{c})$ learns to predict the injected noise by minimising $C(\phi) = \mathbb{E}_{k, \mathbf{q}_0, \eta} \|\eta - \eta_\phi(\mathbf{q}_k, k, \mathbf{c})\|^2$, with $\mathbf{c}$ entering by channel concatenation; scalars are broadcast across time and waveforms are stacked. Inference uses the deterministic DDIM update [12].

We validate on a deliberate hierarchy of systems of increasing complexity. (i) The Ornstein–Uhlenbeck process gives an analytical baseline with closed-form moments and autocorrelation function [13]. (ii) The driven stochastic Duffing oscillator with combined white and coloured noise (given by the Ornstein–Uhlenbeck process) [14] is nonlinear, can be chaotic, and stochastic; it is conditioned on the driving field amplitude. (iii) The magnetic DW oscillator in the collective-coordinate model [15] is phenomenologically Duffing-like [7] and forms a physically grounded RC substrate, conditioned on the full driving field. (iv) An experimental interconnected nanomagnetic ring array [9], driven by a rotating in-plane field, is the canonical spintronic reservoir in our group; no closed-form SDE is available, and conditioning is again the full driving field. Figure 1 shows generated trajectories for each system.

Across all four systems, generated ensembles match the data in time-resolved mean, variance, and autocorrelation function in the held-out conditioning set. For the OU process the surrogate recovers the analytic moments $\mathbb{E}[q(t) | q_0 = 0] = 0$ and $\text{Var}[q(t)] = r(1 - e^{-2\theta t})$ across $r \in [1/30, 3]$. The same U-Net, with only its conditioning channels changed, reproduces the Duffing oscillator, DW oscillator, and nanoring statistics. We further exploit the differentiability of g_ϕ : gradients backpropagated through the DDIM chain recover physical parameters from observed trajectories (e.g., for the Duffing oscillator, the driving field amplitude is recovered with a mean absolute percentage error of $5.74\% \pm 3.9\%$ with 95% of 100 trials below 12.3%), demonstrating gradient-based inverse design on substrates lacking a simulator.

These results establish conditional diffusion models as faithful surrogates for spintronic RC substrates (and, indeed, any dynamical system)—requiring only observed trajectories and no underlying equation of motion—with differentiability opening a route to parameter inference, material design, and optimisation of device networks.

References

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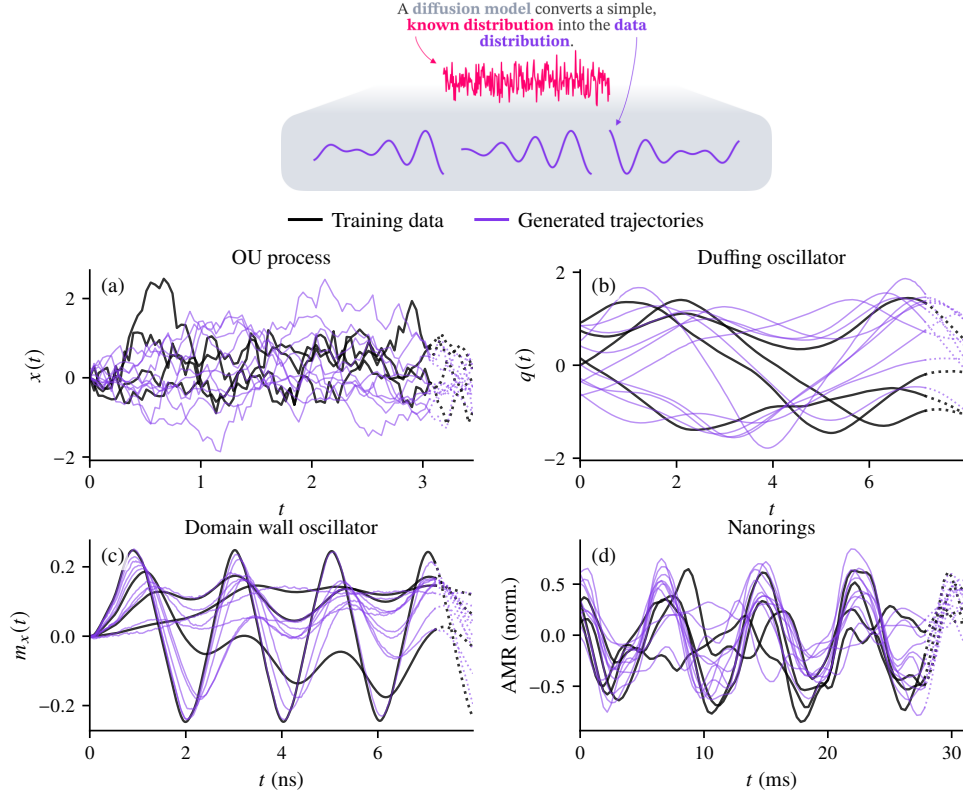


Fig. 1: (top) The diffusion model transforms a known noise distribution into the trajectory distribution of each substrate. (bottom) Generated trajectories (purple, thin) from the held-out conditioning test set provide good agreement with the training data (black, thick) for (a) the OU process, (b) the Duffing oscillator, (c) the DW oscillator, and (d) experimental nanoring AMR signals.

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